IIIT SRI CITY RESEARCH EXPO 2026

RESEARCH PROPOSAL · IN PROGRESS

Predicting Yield

Decoding Nitrogen Control of Yeast with Hybrid Mechanistic–ML Models

Same strain | Same process | Different raw-material lot | **Different yield. Why, and can we predict it?**

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01 The problem: nitrogen decides whether yeast grows or produces

Each lot of yeast extract, peptone or molasses carries a different amount and mix of nitrogen. Yeast senses it (SPS, TORC1), prioritises sources (NCR) and buffers with spare capacity, so yield does not follow nitrogen in a straight line.

>50%
pathway capacity held in reserve under nitrogen limitation

~1%
of theoretical yield reached by most recombinant proteins in yeast

None
of the 20 papers predicts yield from a lot's nitrogen profile

02 The evidence: 20 papers, 5 layers, 4 model families

Sense → Allocate → Quantify → Models → Predict NCR runs through every layer

Kinetic
R^2 > 0.99 on an unseen run
One substrate; no NCR, no reserves

Genome-scale
Enzyme limits: error 129% → 42%
Steady state; secreted protein ~1,000× too high

Machine learning
47–86% lower error
Black box; data-hungry; extrapolates poorly

Hybrid · best fit
Test RMSE 5.60 → 2.87
ML corrects the equations' residual

03 My approach: match nitrogen demand to nitrogen supply

DEMAND · protein DNA sequence

$$D_N = \Sigma\, n_i \cdot a_i$$

amino-acid count × N atoms each (Arg 4, His 3, Lys/Asn/Gln/Trp 2, others 1)

SUPPLY · lot analysis (YAN, forms, C:N)

$$v_j = \varphi_{NCR}(YAN) \cdot V_{max} C_j / (K_m + C_j)$$

each N form → an uptake limit; the NCR switch holds back poor sources

↓

CELL MODEL · existing tools, stacked

$$S \cdot v = 0 \quad v \le k_{cat} \cdot E$$

Metabolism
Yeast9 GEM + GECKO enzyme limits, gaps filled by DLKcat

Secretion and proteome
pcSecYeast: cost of folding and exporting the protein

NCR + reserves (new)
nitrogen form and spare capacity; swapped per host

↓

LEARN · hybrid correction

$$Y_h = Y_m + Y_{RM}$$

ML learns the residual; Monte Carlo gives a confidence band

OUTPUT · for each lot
Yield ± uncertainty · limiting N source · resistance score

$$R = 1 - W_{FVA} / v_{pred}$$

04 Multi-omics: each layer tunes a part of the model

Genomics
gene function; host regulation plug-in

Transcriptomics
prune unexpressed reactions (GIMME, iMAT)

Proteomics
enzyme amounts, reserves, secretion load

Metabolomics
pathway choice; ^{15}N tracing of nitrogen

Structure
AlphaFold check on the product

Gaps, hypothesis and roadmap continue overleaf →

From gaps to a testable model

05 Open gaps, and the layer that closes each one

Gap in current models	My plan	Layer
No model links feedstock nitrogen to yield	Map each nitrogen form in a lot to an uptake limit; predict yield per lot	**Supply**
All nitrogen sources treated as equivalent	NCR switch φ_{NCR}(YAN) delays poor sources until preferred N runs out	**NCR plug-in**
Reserves and NCR not linked to protein synthesis	Reserve limits on enzyme and translation capacity in the cell model	**Reserves**
The product's own nitrogen demand is ignored	Compute it from the DNA sequence: the same lot can suit one protein and not another	**Demand**
Secreted protein overpredicted ~1,000×	Proteome and secretion limits on folding and export (pcSecYeast)	**Secretion**
Mostly steady-state chemostat data	Calibrate on nitrogen-pulse time courses; dynamic kinetic core	**Kinetic**
Mechanistic drift; ML is a black box	ML learns only the residual; validated on held-out nitrogen conditions	**Learn**
Strains and hosts regulate nitrogen differently	Host-specific regulation plug-in, refitted per strain with omics	**NCR plug-in**
Predictions carry no uncertainty	Monte Carlo confidence bands plus an FVA-based resistance score	**Output**

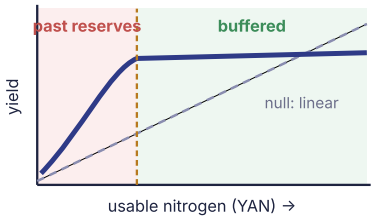
06 Hypothesis, and how it could be proven wrong

> **Yield drops only when a lot's nitrogen pushes cells past their reserve capacity.** Until then, NCR and reserves buffer the change.

H1 Threshold: yield vs YAN is flat, then drops. Wrong if it falls linearly.

H2 Form matters: equal YAN, different N forms → different yield. Wrong if YAN alone predicts it.

H3 Robustness: narrow FVA width → steady titres. Wrong if replicate spread doesn't track it.



Predicted shape, not data

Test: train on public chemostat and nitrogen-pulse data → validate on held-out nitrogen sources → compare RMSE against kinetic-only and GEM-only models → real lots with varied YAN in the lab.

07 Roadmap

NOW
Review and data
Review paper; seven public nitrogen datasets compiled

PHASE 1
Build the core
Demand, supply and cell layers in S. cerevisiae; simulated lots

PHASE 2
Learn and compare
ML residual and uncertainty; benchmark on held-out data

PHASE 3
Validate and extend
Real lots in the lab; a second host such as K. phaffii; live digital twin

08 Why it matters

Screen a raw-material lot on a computer before it reaches the fermenter, instead of weeks of trial runs per lot. Fewer failed batches, consistent titres, and cheaper, variable feedstocks become usable.

Your idea could be the missing piece

Datasets, methods or industry experience with raw-material variability: let's talk.

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